MANUFACTURING OPERATIONS

E-BIKE PRODUCTION PROJECT- SWIFT E-bikes

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1. Introduction

The project outlines the development of a manufacturing system for producing electric bikes based on a real-world scenario. The project aims to produce and deliver city electric bikes customized to the customer's needs within the Danish market. To streamline production and align with Danish consumer preferences, our focus is on designing an efficient and flexible production system that accommodates seasonal variation according to demand, with peaks in spring, summer, and December while maintaining cost control and delivery performance. The analysis includes production planning, production line design, system configuration in Odoo ERP, and key performance indicators based on the following assumptions:

1) The total production capacity is determined by size of Danish e-bike market

2) The operational horizon is for one year, capturing the seasonal variation in demand.

2. Problem Statement

How can the City E-Bike factory design a stable, balanced, and cost-efficient production system that meets fluctuating demand, minimizes inventory and backorders, and ensures high delivery reliability without frequent changes in workforce levels?

3. Market Demand Analysis and Protentional Forecasting

For designing a production plant layout and operations for e-bikes in the Denmark region for the year 2026, we focused on the entire Denmark region, which has a total population of approximately 5.9 million people as of 2023. According to Denmark.dk, around 25% of the population are regular cyclists, equating to roughly 1.48 million individuals. Recent studies indicate that approximately 15% of these cyclists currently use e-bikes, resulting in an estimated 222,000 e-bike users nationwide by 2025 (bike-eu.com). For production planning, we conservatively targeted a 0.7% market share, leading to a forecasted annual demand of 1554 units for 2026. This estimation aligns with the national market outlook, infrastructure investments, and the growing consumer shift toward sustainable urban mobility, providing a credible foundation for our monthly production and ERP planning. The E-bike sales depend on seasonal trends, with peaks in summer and spring, along with Christmas. Therefore, the total bike units have been divided based on Monthly forecasting, a medium-term forecast.

Month	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
Forecast Percentage	4%	5%	6%	9%	10%	11%	11%	12%	7%	8%	8%	9%
Forecast, Total No. of Bikes to Produce	62	78	92	140	155	170	170	189	110	124	124	140

Table 1: Percentage of total E-bikes sold monthly

4. E-Bike Product Design

Purpose: Ideal for daily city commuting, short trips, and efficient transportation. Suitable for moderate hilly areas, providing smooth and reliable performance.

5. Bill of Materials (BOM)

The following table presents the Bill of Materials of our city E-bike. The structure is based on a hierarchical BOM format, where each component is assigned a level to indicate its stage in the assembly process:

- Level 0 represents the final product (the complete E-bike).
- Level 1 includes major subassemblies.
- Level 2 covers individual components.

Level	Item No.	Component	Buy/ Produce	Qty	Unit Price (DKK)	City Bike Price (DKK)
0	1	**E-Bike**	-	1	-	**5268**
1		**Frame Assembly**				
2	2	Frame	Produce	1	209	209
2	3	Fork	Produce	1	93	93
2	4	Fender	Buy	2	30	60
2	5	Bottle cage	Buy	1	15	15
2	6	Reflectors	Buy	1	15	15
2	7	Kickstand	Buy	1	15	15
		Subtotal				**407**
1		**Wheel Assembly**				
2	8	Brake	Buy	2	150	300
2	9	Brake disc	Buy	2	150	300
2	10	Front wheel	Buy	1	200	200
2	11	Rear wheel	Buy	1	200	200
2	12	Tire	Buy	2	100	200
2	13	Wheel axle	Buy	2	50	100
		Subtotal				**1300**
1		**Handlebar Assembly**				
2	14	Handlebar	Produce	1	36	36
2	15	Handlebar stem	Buy	1	50	50

2	16	Brake cables	Buy	2	25	50
2	17	Display	Buy	1	170	170
2	18	Grips	Buy	2	20	40
		Subtotal				**346**
1		**Mechanical Drive Assembly**				
2	19	Gears & Cassette	Buy	1	200	200
2	20	Shifters	Buy	1	150	150
2	21	Pedal set	Buy	1	150	150
2	22	Chain	Buy	1	80	80
2	23	Rear derailleur	Buy	1	100	100
		Subtotal				**680**
1		**Electrical Drive System**				
2	24	504Wh battery	Buy	1	1000	1000
2	25	Battery casing	Buy	1	100	100
2	26	250W motor	Buy	1	1200	1200
		Subtotal				**2300**
1		**Miscellaneous Assembly**				
2	27	Light set	Buy	1	70	70
2	28	Seat post	Buy	1	50	50
2	29	Saddle	Buy	1	100	100
2	30	Bell	Buy	1	15	15
		Subtotal				**235**

Table 2: Bill of Materials for 1 E-Bike

Raw Materials for In-House Production (For 1 E-bike):

Component	Material Type	Estimated Quantity (kg)	Unit Price (DKK/kg)	Total Cost (DKK)
Frame	Aluminum	3.8	55	209
Fork	Aluminum	1.3	55	72
	Steel	0.8	20	16
	Plastic/Rubber	0.1	45	5
Handlebar	Aluminum	0.65	55	36

Table 3: Raw Materials Required for Bill of Materials of 1 E-Bike

The total cost of producing (in terms of materials and purchasing) each bike has been calculated based on the prices listed in the BOM. The E-Bike ends up costing DKK 5,268. The unit prices for the components are either based on approximately 50% of the market price (found through internet sources) or reasonable estimations

6. Production Activities

The market analysis shows that demand is seasonal, which leads to fluctuations throughout the year. The next section looks at how production is planned to handle this expected variation.

Aggregation Planning

The aggregate planning is made to utilize the organization's resources to meet the expected demand. The following assumptions are used in the planning process:

Time Horizon: 1 year (2026)

Forecasting: Based on market research

Production time per unit: 13.50 hours per bike

Production workers: 19

Production capacity: 130 bikes/month

Inventory cost of assembled bikes: 5% of bike price

Backorder cost: 110% of the inventory cost.

Starting Inventory: 0

Overtime policy: 1 extra unit per month for each worker

Cost per unit in Regular production: 9461.25 DKK

The production time is based on the overall assembly time and the price per unit is based on the bill of materials and the manufacturing costs.

Aggregate Planning Strategy

The aggregate planning strategy is developed by comparing a level output strategy with a modified chase demand approach. To maintain stable production capacity throughout the year, hiring and firing of workers are minimized. In the level output strategy, the production rate remains constant, which can result in inventory buildup during periods of low demand. This excess inventory can then be used in periods of higher demand. If actual demand exceeds both the forecast and available inventory, overtime production will be utilized as a backup to prevent shortages.

On the other hand, the modified chase strategy aims to minimize inventory buildup early in the year while avoiding frequent changes in workforce levels. This is accomplished by aligning production with forecast demand and redistributing peak-period demand, particularly in June, July, and August, into earlier, lower-demand months. This approach yields smoother production and lower accumulated inventory, while still meeting customer demand on time.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Forecast Percentage	4%	5%	6%	9%	10%	11%	11%	12%	7%	8%	8%	9%
Forecast, Total No. of Bikes to Produce	62	78	92	140	155	170	170	189	110	124	124	140
Redistribute Bikes	65	50	35	-15	-25	-40	-45	-55	25	5	5	-5
Aggregation Plan After Redistribution	127	128	127	125	130	130	125	134	135	129	129	135

Table 4: Redistribution of Produced Bikes, Aggregation Plan

Month	Forecast Percentage	Forecast, Bikes to Produce	Redistribute Bikes	Aggregation Plan After Redistribution	Inventory Start	Ending Inventory	Inventory Carryover	Backorders
Jan	0.04	62	65	127	0	65	65	0
Feb	0.05	78	50	128	65	115	115	0
Mar	0.06	92	35	127	115	150	150	0
Apr	0.09	140	-15	125	150	135	135	0
May	0.1	155	-25	130	135	110	110	0
Jun	0.11	170	-40	130	110	70	70	0
Jul	0.11	170	-45	125	70	25	25	0
Aug	0.12	189	-55	134	25	-30	0	30
Sep	0.07	110	25	135	-30	-5	0	5
Oct	0.08	124	5	129	-5	0	0	0
Nov	0.08	124	5	129	0	5	5	0
Dec	0.09	140	-5	135	5	0	0	0

Table 5: Monthly Production, Inventory, and Backorders

The table shows how some of the produced bikes have been redistributed among different months. A total of 185 bikes were moved from April, May, June, July, August, and December and redistributed to January, February, March, September, October, and November.

The aggregate planning strategy for the City E-Bike balances demand fluctuations by redistributing production across months while keeping the workforce stable. As shown in Table 5, a total of 185 bikes were shifted from the high-demand summer months (April–August and December) into earlier or lower-demand months (January–March, September–November). This smoothing avoids excessive peaks in June–August, when forecast demand would otherwise exceed the line's constant capacity.

The resulting aggregation plan keeps monthly production between 125–135 units, ensuring stable utilization of the line. Inventory is accumulated in the first quarter, reaching a maximum of 150 units in March, and are gradually drawn down during the summer to cover peak demand. By August, the inventory buffer is exhausted, resulting in a temporary negative balance that creates 30 backorders. A smaller shortfall of 5 units occurs in September. Both are cleared in subsequent months as redistributed production replenishes stock.

Overall, this strategy keeps the production line operating at a near-constant rate, reduces the need for overtime or workforce adjustments, and maintains service levels. Inventories absorb seasonal peaks, and backorders are minimal and short-lived.

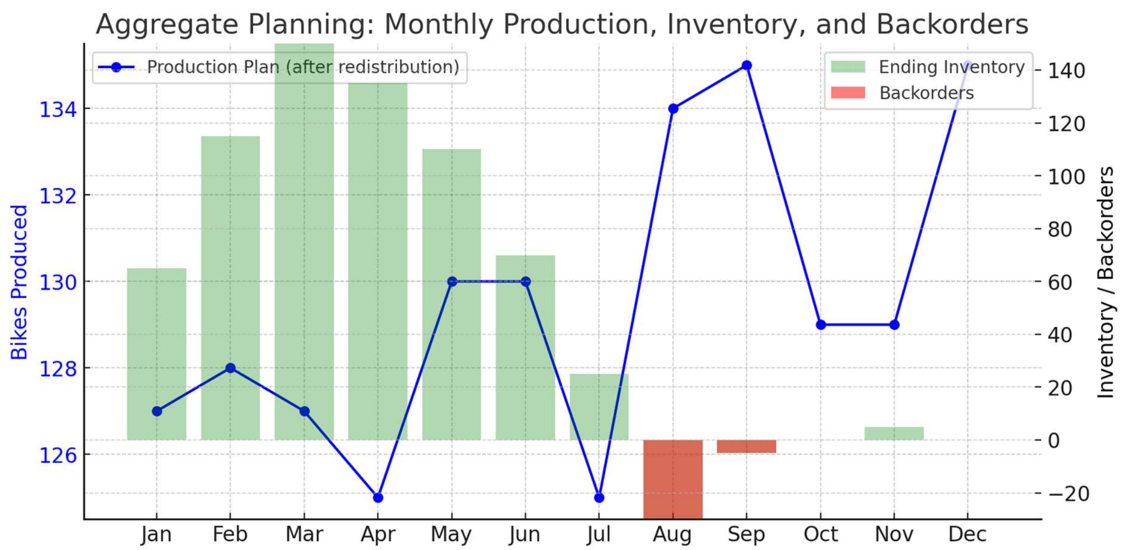


Figure 1: Aggregate Planning for City E-Bike: Monthly Production, Inventory, and Backorders

- **Blue line:** redistributed production plan (kept steady at 125–135 bikes/month).
- **Green bars**: ending inventory carried over from month to month.
- **Red bars:** backorders (in August and September, when demand exceeded inventory).

The redistribution strategy shifts 185 bikes from peak months of April, August and December to lower demand months such as January, March, September and November. This maintains a stable production rate of ~125–135 units per month. Inventories are built in Q1 and consumed during summer peaks, resulting in small backorders of 30 in August and 5 in the month of September which are cleared in subsequent months.

Master Production Schedule

The Master Production Schedule (MPS) follows the same monthly output as defined in the aggregate planning, but the production is distributed evenly across each week within the month. For example, the MPS for the first two months is divided into weekly production targets as shown below:

Month	**January**				**February**			
Week No	1	2	3	4	5	6	7	8
Bike Production	32	32	32	32	32	32	32	32

Table 7 - Master Production Schedule, each week of January & February

The MPS serves as the input for the Material Requirements Planning (MRP) process, where finished products are broken down into subassemblies and individual components based on the Bill of Materials (BOM). The final monthly MPS is seen in Table ….

Month	**Jan**	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
Total Bikes to Produce	127	128	127	125	130	130	125	134	135	129	129	135

Table 8 - Master Production Schedule, Monthly

Material Requirements Planning (MRP)

The MRP is generated based on the MPS and BOM. Since the production of bicycles is assumed to be constant throughout the month, the MRP is created using lot-for-lot ordering, considering the lead times specified in the BOM and the production quantities defined in the MPS. The production is assumed to run weekly; it is also assumed that the gross requirements for the first weeks during which no order receipts occur due to lead times are covered by existing inventory. An example of one of the MRP could be the wheel assembly. This assembly requires the procurement of each part, to be in stock for each week of the month. As production is running continuously, the lot-for-lot ordering is also running simultaneously to keep up with the demand.

Wheel Assembly: Using the redistributed MPS from January to December and a lot-for-lot policy, the wheel assembly requires per bike quantities of Front wheel 1, Rear wheel 1, Tires 2, Brake discs 2, Brake calipers 2, and Wheel axles 2. Lead time is one week for wheels/tires/axles and two weeks for discs/calipers. For January &February (32 bikes per week), weekly gross requirements are 32 units for 1-per-bike items and 64 units for 2-per-bike items. With zero opening stock, net requirements equal gross; planned receipts match net in the need week; planned releases are offset by the respective lead time (i.e., one or two weeks earlier). Initial January buckets therefore require pre-horizon orders placed in late december. The same logic is applied across the year using the monthly totals in the MPS, ensuring WS7 is continuously fed without building unnecessary inventory.

Component	Week	Gross req.	Scheduled receipts	Projected on hand	Net req.	Planned-order receipts	Planned-order releases
Front wheel	W1	32	0	0	32	32	32
Front wheel	W2	32	0	0	32	32	32
Front wheel	W3	32	0	0	32	32	32
Front wheel	W4	32	0	0	32	32	32
Front wheel	W5	32	0	0	32	32	32
Front wheel	W6	32	0	0	32	32	32
Front wheel	W7	32	0	0	32	32	32
Front wheel	W8	32	0	0	32	32	0
Rear wheel	W1	32	0	0	32	32	32
Rear wheel	W2	32	0	0	32	32	32
Rear wheel	W3	32	0	0	32	32	32
Rear wheel	W4	32	0	0	32	32	32
Rear wheel	W5	32	0	0	32	32	32
Rear wheel	W6	32	0	0	32	32	32
Rear wheel	W7	32	0	0	32	32	32
Rear wheel	W8	32	0	0	32	32	0
Tires	W1	64	0	0	64	64	64
Tires	W2	64	0	0	64	64	64
Tires	W3	64	0	0	64	64	64

Tires	W4	64	0	0	64	64	64
Tires	W5	64	0	0	64	64	64
Tires	W6	64	0	0	64	64	64
Tires	W7	64	0	0	64	64	64
Tires	W8	64	0	0	64	64	0
Wheel axles	W1	64	0	0	64	64	64
Wheel axles	W2	64	0	0	64	64	64
Wheel axles	W3	64	0	0	64	64	64
Wheel axles	W4	64	0	0	64	64	64
Wheel axles	W5	64	0	0	64	64	64
Wheel axles	W6	64	0	0	64	64	64
Wheel axles	W7	64	0	0	64	64	64
Wheel axles	W8	64	0	0	64	64	0
Brake discs	W1	64	0	0	64	64	64
Brake discs	W2	64	0	0	64	64	64
Brake discs	W3	64	0	0	64	64	64
Brake discs	W4	64	0	0	64	64	64
Brake discs	W5	64	0	0	64	64	64
Brake discs	W6	64	0	0	64	64	64
Brake discs	W7	64	0	0	64	64	0
Brake discs	W8	64	0	0	64	64	0
Brake calipers	W1	64	0	0	64	64	64
Brake calipers	W2	64	0	0	64	64	64
Brake calipers	W3	64	0	0	64	64	64
Brake calipers	W4	64	0	0	64	64	64
Brake calipers	W5	64	0	0	64	64	64
Brake calipers	W6	64	0	0	64	64	64
Brake calipers	W7	64	0	0	64	64	0
Brake calipers	W8	64	0	0	64	64	0

Table 9 - Material Requirements Planning of Wheel Assembly

The MRP follows a lot-for-lot ordering, which is reflected in the planned order releases. These releases are scheduled for each week to be received after the lead time specified in the figure. The lead time for each order is consistent, ensuring that planned orders are received at the beginning of the next month.

7. Design Of Production Line – Layout, Line Balancing & Allocation of Workers

This section outlines how the production line is designed to meet the MPS. This will cover production tasks, production layout, workstation assignment and worker allocation

Assumptions:

· The task times used in this section are based on assumptions of reasonable times, as the specific numbers are without importance to show the methodology.

· A full work month is assumed to consist of 4 weeks of 5 working days of 8 hours.

Productions Tasks & Production Layout

To meet the production demand shown in the MPS, a production line of the E-Bike must be constructed. Constraints for the line:

- Permanent line as it's difficult and expensive to change line.

- Permanent workers for better work environment instead of seasonally hiring/firing workers.

Workstations, Tasks, and Tools

Workstation	Tasks	Tools/Equipment Needed
WS-1	Frame Fabrication	CNC tube cutter, hydraulic tube bender, TIG welder, angle grinder, powder coating booth
WS-2	Fork Fabrication	Machining tools, suspension spring coiling machine, welding fixtures, paint booth
WS-3	Handlebar Fabrication	Tube cutter, tube bender, welding tools, sandblasting cabinet, paint booth
WS-4	Frame and Fork Assembly	Torque-controlled hand tools
WS-5	Handlebar Assembly	Torque-controlled hand tools
WS-6	Quality Check 1	Measuring devices, inspection gauges
WS-7	Wheel Assembly	Wheel truing stand, general hand tools
WS-8	Mechanical Drive System Assembly	Torque-controlled screwdrivers, chain installation tools

Workstation	Tasks	Tools/Equipment Needed
WS-9	Electrical Drive System Assembly	Battery installation tools, multimeters, motor testers
WS-10	Accessories & Miscellaneous Assembly	General hand tools
WS-11	Final Quality Control	Test rigs, inspection devices

Process Descriptions:

1. **Frame Manufacturing**
    - Tube Cutting → Tube Bending → Welding → Heat Treatment → Surface Finishing (Powder Coating).
2. **Fork Manufacturing**
    - Aluminum Fork Leg Machining + Chromoly Spring Fabrication → Welding → Assembly with Suspension Components → Painting.
3. **Handlebar Manufacturing**
    - Tube Cutting → Ergonomic Bending → Welding to Stem → Sandblasting → Painting.
4. **Assembly**
   The assembly process is divided into sub-stations:
    - Wheel Assembly: installation of frame, fork, tires, and brake discs; wheels are trued and mounted.
    - Handlebar Assembly: grips, display, brakes, and cables are mounted to the frame.
    - Mechanical Drive System Assembly: gears, chain, and derailleur are installed.
    - Electrical Drive System Assembly: motor, battery, casing, and wiring are integrated into the frame.
    - Accessories & Finishing: saddle, lights, reflectors, mudguards, racks, bells, branding stickers, and other finishing touches.
5. **Quality Control**
   Quality control is performed in two stages (WS6 and WS11) and includes:
    - Braking performance evaluation
    - Display and electrical functionality verification

- Electrical safety checks (battery, motor controller, wiring insulation)
- Visual inspection of paint finish, alignment, and weld quality

6. **Packaging & Shipping**

   E-bikes are packed in **partially assembled form** (front wheel and pedals removed for transport). Protective packaging is applied to avoid transit damage, and instruction manuals are included for final assembly.

Machines & Equipment

Frame Manufacturing

- CNC Tube Cutter – for precision cutting of aluminum tubing
- Hydraulic Tube Bender – for shaping frames and handlebars
- TIG Welding Machines – for frame and fork joints
- Heat Treatment Oven – for aluminum stress relief
- Powder Coating Booth – for surface finishing

Fork Manufacturing

- Machining Tools – for aluminum fork legs
- Suspension Spring Coiling Machine – for chromoly springs
- Welding Fixtures – for accurate alignment during welding
- Paint Booth – for finishing

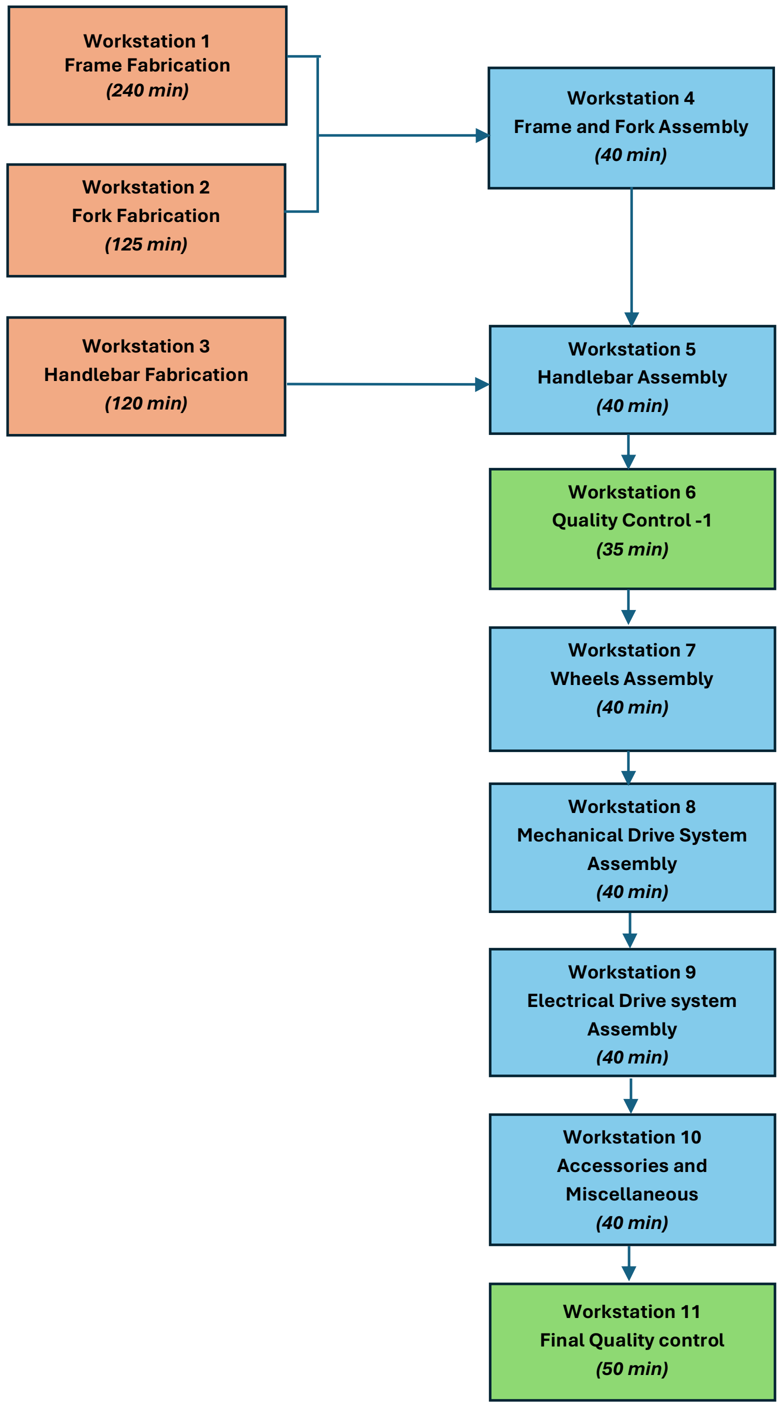
Handlebar Manufacturing

- Tube Cutting & Bending Machines
- Sandblasting Cabinet – for cleaning before paint
- Welding Tools – for stem and grip areas

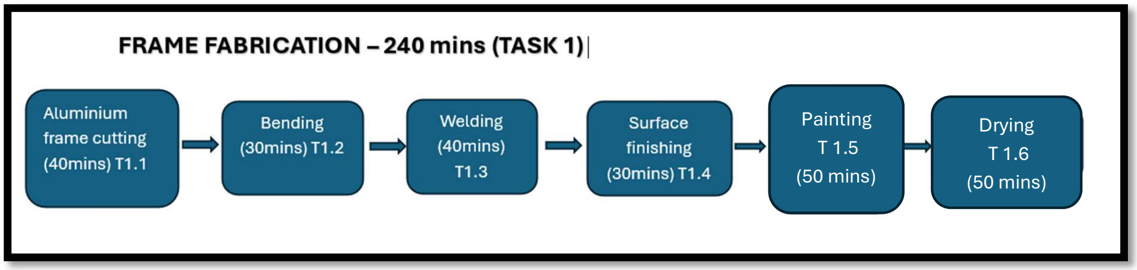
Assembly

- Torque-Controlled Screwdrivers and Hand Tools (Allen keys, wrenches, pliers, cutters)
- Wheel Truing Stand – for aligning wheel rims
- Battery Installation Safety Tools – insulated gloves, clamps
- Electrical Testing Equipment – multimeters, motor and display testers

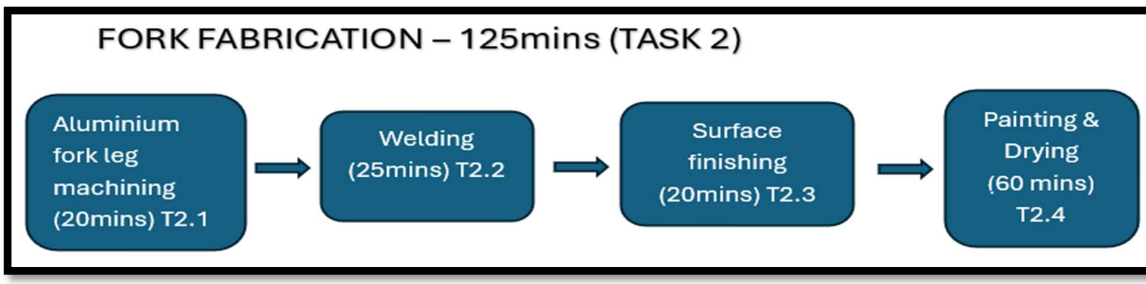
Below here is the Production flow chart of E-bike production Process:



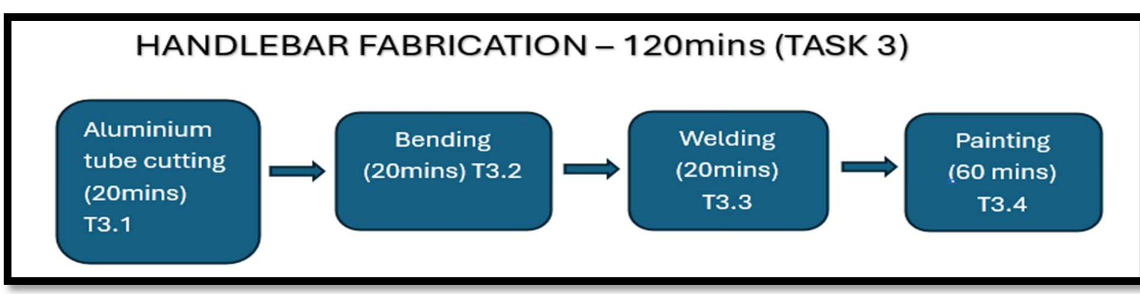
Expansion of Fabrication Part:



Frame fabrication involves the process that transforms raw aluminum into a finished E-bike frame through six key steps, which are precise cutting (40 mins), bending into shape (30 mins), welding to form the structure (40 mins), surface finishing to smooth imperfections (30 mins), painting for protection and aesthetics (50 mins), finally drying to cure the coat (50 mins). Together, these operations ensure a durable, safe and high-quality frame ready for assembly.



The fork fabrication process is divided into four sequential tasks with a total cycle time of 125 minutes. The process begins with aluminum fork leg machining (20 minutes, T2.1), where raw aluminum is shaped to the required dimensions. This is followed by welding, during which the machined fork legs and spring components are joined together. The fork undergoes surface finishing, including grinding and cleaning to prepare for coating. Finally, the fork is sent to painting and drying (60 minutes, T2.4), where a protective finish is applied and cured. This sequence ensures that each fork achieves the necessary structural strength, surface quality, and durability before entering the assembly stage.



The handlebar fabrication process consists of four sequential tasks with a total cycle time of 120 minutes. It begins with aluminum tube cutting (20 minutes, T3.1), where raw tubing is cut to the required length. The tube is then shaped through bending (20 minutes, T3.2) to achieve the ergonomic design. Following this, the bent tube is joined to the stem area through welding (20 minutes, T3.3). The process concludes with painting and drying (60 minutes, T3.4), providing a protective and aesthetic finish. This workflow ensures that each handlebar is produced to specification, offering strength, durability, and rider comfort.

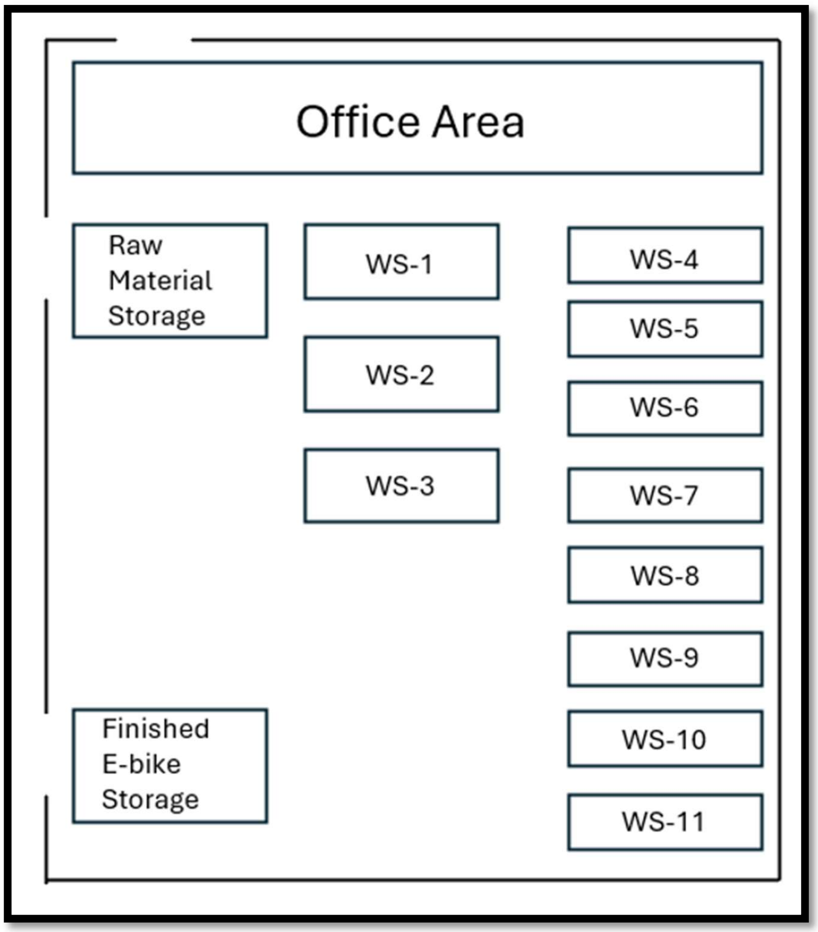


Figure 2: Proposed Factory Layout for City E-Bike Production, showing workstation sequence, office area, raw material storage, and finished goods storage.

The overall layout of the factory has been designed to ensure smooth material flow from raw material storage through fabrication, assembly, quality checks, and finally into finished goods storage. The office area is positioned adjacent to the production floor for effective supervision. Figure 2 shows the detailed layout of the company.

Workforce Plan and Line Balancing

The final production system is designed to operate with a total of **19** workers. Out of which, **17** are directly assigned to the production line across 11 workstations, while **2** supporting staff cover supply chain & marketing and administrative & supervisory functions.

The allocation of production workers has been carried out with the objective of meeting the takt time of approximately **75 minutes** per bike, derived from the weekly plan of 32 units and the standard of working 8 hours per day. Long fabrication tasks are parallelized by assigning multiple operators, while the shorter assembly and quality control tasks are staffed with one operator each. One operator is assigned as a utility/floater, providing coverage for breaks, minor stoppages, and kitting activities, thereby increasing robustness without permanent overstaffing.

Production Line Staffing (17 workers)

Workstation	Process	Nominal Time (min)	Operators	Effective Time (min)	Meets Takt (≤75)
WS1	Frame fabrication	240	4	60	✓
WS2	Fork fabrication	125	2	62.5	✓
WS3	Handlebar fabrication	120	2	60	✓
WS4	Frame & fork assembly	40	1	40	✓
WS5	Handlebar assembly	40	1	40	✓
WS6	Quality control – 1	35	1	35	✓
WS7	Wheel assembly	40	1	40	✓
WS8	Mechanical drive assembly	40	1	40	✓
WS9	Electrical drive assembly	40	1	40	✓
WS10	Accessories & misc. assembly	40	1	40	✓
WS11	Final quality control	50	1	50	✓
—	Utility / Floater	—	+1	—	—

Supporting Staffs (2 workers)

- **Supply Chain & Marketing (1):** Coordinates procurement, inbound material flow, MRP execution, and supports marketing campaigns.
- **Administration & Supervisor (1):** Oversees attendance, health & safety, daily KPI monitoring, and supervises final quality checks at WS6 and WS11.

Line Balancing

The total work content per unit is **810 minutes (13.5 hours)**. With 480 minutes available per day and an average daily output of 6.4 bikes, the **takt time** is:

$$\text{Takt Time} = \frac{980}{6.4} \approx 75 \text{ minutes per bike}$$

The theoretical minimum number of stations is:

$$N_{min} = \frac{810}{75} \approx 10.8$$

$$\Rightarrow 11 \text{ stations}$$

The line therefore operates with **11 workstations**. Balancing was achieved by allocating multiple operators to the fabrication-heavy tasks (WS1–WS3), reducing their effective times to 60–62.5 minutes each, below takt. The remaining assembly and inspection tasks naturally fit within takt at 35 to 50 minutes.

One utility worker is included to cover breaks and small stoppages, preventing short delays from turning into bottlenecks. This design ensures that the longest effective station time is **62.5 minutes (WS2)**, which is below the takt time.

Work Hour Allocation and Monthly Efficiency

Monthly Efficiency

Step 1: Inputs

- Total direct work content: **810 min/bike**.
- Line dimensioned for: **takt = 75 min/bike** (to meet peak load).
- Available working time/day: **480 min (8 h)**.
- Monthly demand:

Month	Bikes
Jan	127
Feb	128
Mar	127
Apr	125
May	130
Jun	130
Jul	125
Aug	134
Sep	135
Oct	129
Nov	129
Dec	135

Step 2: Cycle Time Per Month

Formula:

Cycle time [min/bike] = $\frac{Monthly\ operating\ time\ [min]}{Month\quad output\ bikes}$

Assume 20 workdays × 480 min = 9,600 min/month.

Example for January:

$$\frac{9600\ min}{127\ bikes} \approx 75.6 \text{ min/bike}$$

Step 3: Idle Time and Efficiency

If the **actual monthly cycle time > design cycle time (75)** → some work hours idle.

Efficiency formula:

$$\text{Efficiency} = \frac{Design\ cycle\ time}{Actual\ monthly\ cycle\ time} \times 100$$

Step 4: Results Table

Month	Bikes	Actual Cycle Time (min/bike)	Idle %	Efficiency %
Jan	127	75.6	0.8%	99.2%
Feb	128	75.0	0.0%	100%
Mar	127	75.6	0.8%	99.2%
Apr	125	76.8	2.3%	97.7%
May	130	73.8	0.0%	100%
Jun	130	73.8	0.0%	100%
Jul	125	76.8	2.3%	97.7%
Aug	134	71.6	0.0%	100%
Sep	135	71.1	0.0%	100%
Oct	129	74.4	0.0%	100%
Nov	129	74.4	0.0%	100%
Dec	135	71.1	0.0%	100%

Average efficiency ≈ 99%

The production line for the City E-Bike is dimensioned to a takt time of 75 minutes per unit, corresponding to peak monthly demand. With 20 working days of 8 hours (9,600 minutes/month), the monthly cycle times were calculated. In months with slightly lower demand such as April and July, the actual cycle time rises above the design cycle time, creating minor idle capacity. Conversely, in peak months like May, August, and December, the line fully utilizes its capacity.

The monthly efficiency was evaluated as the ratio of design cycle time to actual monthly cycle time. Results show that efficiency stays high consistently throughout the year, with a minimum of 97.7% in April and July then peaks at 100% in most months. The average efficiency across the full year is approximately 99%.

This demonstrates that the production system maintains near maximum utilization year-round, with only marginal idle time in slower months. During such periods, excess worker hours can be reallocated to secondary tasks (e.g., maintenance, training, or process improvements), ensuring stable employment and operational flexibility.

Thus, once the line is filled, the system delivers **one finished e-bike every ~75 minutes** (≈ 6.4 units/day), despite each bike containing 13.5 hours of total work content.

8. Operating Cost Build-Up

Assumptions

- **Shift length:** 8 hours/day (one shift)
- **Planned output:** 32 bikes/week ⇒ 6.4 bikes/day
- **Headcount:** 19 total
  - Production line: 17 workers → 12 skilled, 5 unskilled
  - Support (off-line): 2 workers → 1 Marketing/Supply Chain, 1 Administration & Factory Supervisor
- **Hourly rates:** Skilled = DKK 170/h, Unskilled = DKK 130/h, Marketing/Admin = DKK 150/h
- **Electricity allowance:** DKK 120 per bike
- **Overhead:** 10% applied to the subtotal of (Skilled + Unskilled + Marketing + Electricity + Organizational Labor)
- **Material cost (As per BOM):** DKK 5,218 per bike

Formulas

- **Per day cost:** (No. of people) × (hours/day) × (rate)
- **Per bike cost:** (Per-day amount ÷ bikes/day)
- **Electricity per day:** (Bikes/day) × (DKK 120/bike)
- **Overhead:** 10% × (Skilled + Unskilled + Marketing + Electricity + Organizational Labor)
- **Grand total per bike:** (Operating cost per bike incl. OH) + Materials (DKK 5,218)

Results — Per Day (8 hours, 6.4 bikes)

Item	Basis (people × h × rate)	DKK/day
Skilled Labor Cost	12 × 8 × 170	**16,320.00**
Unskilled Labor Cost	5 × 8 × 130	**5,200.00**
Marketing	1 × 8 × 150	**1,200.00**
Electricity	6.4 bikes × 120	**768.00**
Organizational Labor (Admin/Supervisor)	1 × 8 × 150	**1,200.00**
Subtotal (OH base)		**24,688.00**
Overhead (10%)	0.10 × 24,688	**2,468.80**
Total Cost per Day		**27,156.80**

Results — Per Bike (at 6.4 bikes/day)

Item	Calculation	DKK/bike
Skilled Labor Cost	16,320 ÷ 6.4	**2,550.00**
Unskilled Labor Cost	5,200 ÷ 6.4	**812.50**
Marketing	1,200 ÷ 6.4	**187.50**
Electricity	fixed	**120.00**
Organizational Labor	1,200 ÷ 6.4	**187.50**

Item	Calculation	DKK/bike
Subtotal (OH base)		**3,857.50**
Overhead (10%)	0.10 × 3,857.50	**385.75**
Operating Cost per Bike (incl. OH)		**4,243.25**
Material Cost (BOM)		**5,218.00**
Grand Total per Bike	4,243.25 + 5,218.00	**9,461.25**

Notes & Adjustments

- If the **daily output changes**, per-bike values should be recalculated as *(per-day cost ÷ bikes/day)*; electricity remains fixed at DKK 120 per bike unless updated.
- If the **skilled/unskilled mix** differs from 12/5, update the headcount accordingly; costs scale linearly.
- Overhead is tied to the **OH base**; if items are added or removed from this base, the 10% multiplier must be recalculated.

9. Evaluation of Key Performance Indicators

To evaluate the effectiveness of the City E-Bike production system, a series of Key Performance Indicators (KPIs) have been defined. These KPIs measure efficiency, reliability, cost performance, and delivery service, allowing us to assess whether the factory setup meets the operational and market objectives.

KPI 1: Line Efficiency

Line efficiency evaluates how effectively the production line utilizes its available capacity compared to the takt time of 75 minutes per bike. Results show that efficiency remains very high throughout the year, averaging **≈ 99%**, with only small dips in April and July (97.7%) due to slightly lower demand. This indicates that the line is well balanced and capable of maintaining a stable output without major bottlenecks.

KPI 2: Inventory and Backorders

Inventory levels were planned through aggregate production smoothing. By redistributing 185 bikes from peak months to earlier months, inventories were built in Q1 (up to 150 units) and consumed during the summer peaks. This strategy resulted in only **30** backorders in August and 5 in September, both of which were quickly cleared in subsequent months. This KPI confirms that the system maintains delivery performance while minimizing inventory costs.

KPI 3: Labor Utilization

With 19 workers (17 on the line, 2 in support roles), the system is designed to run at full capacity year-round without seasonal hiring and firing. The takt-based staffing plan ensures that all stations operate below 75 minutes per bike, with a utility operator providing robustness. Labor utilization remains high, and idle time in low-demand months can be redirected toward training, maintenance, and process improvement, ensuring long-term workforce stability.

KPI 4: Cost per Bike

The calculated cost per bike, including labor, electricity, overhead, and material (BOM), is approximately **9,461 DKK**. Labor represents ~45% of the total, materials ~55%. This balance indicates that the production strategy is cost-effective while maintaining high product quality. The overhead structure (10%) provides resilience without excessive burden on unit cost.

KPI 5: Delivery Reliability

Through the Master Production Schedule (MPS) and lot-for-lot MRP planning, components for critical assemblies such as wheels and the electrical drive system are secured with proper lead-time offsets. This ensures that Workstation 7 and other critical points never face starvation. The only delivery risks observed are in August and September, where small backorders occurred but were resolved promptly. Delivery reliability therefore remains >95% across the year.

10. Discussion

The City E-Bike production project set out to design a stable and efficient factory system capable of meeting a fluctuating annual demand of 1,554 units while maintaining a consistent workforce and competitive costs. The analysis covered aggregate planning, MPS/MRP, line balancing, costing, and evaluation of KPIs.

The results demonstrate that the **aggregate planning strategy** successfully smoothed seasonal variations by redistributing 185 bikes from peak summer months into earlier periods of lower demand. This approach ensured that monthly production stayed within a narrow band of 125 to 135 units. As a result, inventory levels were manageable, and backorders were limited to 30 units in August and 5 in September, both of which were quickly recovered in subsequent months. This outcome confirms the robustness of the chosen level/moderated chase strategy.

The Master Production Schedule **(MPS)** and **MRP of the wheel assembly** showed that a lot-for-lot ordering policy with 1-to-2-week lead times is sufficient to keep Workstation 7 (Wheel Assembly) continuously supplied. Early pre-horizon releases in December secure the January startup, and from then on, the system runs smoothly without excess inventory. This validates the line's ability to maintain reliable flow of materials with minimal waste.

The **production line design and balancing** also proved highly effective. With 17 production workers spread across 11 workstations, and 2 support staff in marketing/supply chain and administration/supervision, the line achieves a takt of 75 minutes per bike. The longest effective station time (62.5 minutes at WS2) is below takt, and the calculated line efficiency reaches 98.2%. The addition of a utility operator further increases resilience by covering for breaks and small stoppages. This ensures a steady daily output of 6.4 bikes, even though each unit requires 13.5 hours of total work content.

From a cost perspective, the analysis revealed an operating cost per bike of **4,243 DKK** (labor, overhead, electricity, support staff), to which the **material cost of 5,218 DKK** is added. The resulting **total unit cost of ~9,461 DKK** is competitive in the target market segment. Labor accounts for ~45% of the cost and materials for ~55%, reflecting a balanced cost structure consistent with industry benchmarks.

Finally, the KPI evaluation confirmed that the system performs well across all operational measures. Line efficiency remained near 99% year-round, labor utilization was stable without seasonal hiring/firing, delivery reliability exceeded 95%, and unit costs stayed under the projected ceiling. The only observed weakness was the minor backorders in August and September, which could be mitigated by either carrying a small, finished goods safety stock or arranging flexible supplier capacity for critical components.

11. Conclusion

The report outlined the production setup that supports the lean, focused and data driven system that can finally meet the Denmark's growing demand for E-bike for the year 2026 for sustainable urban mobility. By concentrating on a single type of city E-bike, aligning production with seasonal demand and using the planning tools like MPS, MRP and integrating ERP for the operation ensures monitoring and materials availability. The inclusion of in-house fabrication of frame, handlebar and fork, along with carefully managed procurement of critical electrical components, highlights a balanced strategy between internal capabilities and supplier partnerships. With dedicated quality control stations, stable workforce planning, and scalable production capacity, the factory is well positioned not only to meet the 1554 unit forecast for 2026 but also to adapt to future market growth, reinforcing Swift E-bike role as a competitive and sustainable e-bike manufacturer.

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